ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ECE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Compute 8-point DFT using Radix-2 Decimation-in-Time (DIT) FFT algorithm. [14 Marks]

[OR]

Q.1 (Alt) Derive butterfly signal flow graph for Radix-2 DIF FFT algorithm. Circular convolution. [14 Marks]

--- UNIT II ---

Q.2 Design analog Butterworth low-pass filter. Bilinear Transformation and Frequency Warping. [14 Marks]

[OR]

Q.2 (Alt) Design digital IIR filter using Impulse Invariance method from analog transfer function H(s). [14 Marks]

--- UNIT III ---

Q.3 Discuss linear phase FIR filter design using Hamming and Hanning windows. [14 Marks]

[OR]

Q.3 (Alt) Explain Frequency Sampling method for designing FIR filters with linear phase equations. [14 Marks]

--- UNIT IV ---

Q.4 Realize digital filter H(z) using Direct Form-I, Direct Form-II, Cascade, and Parallel. [14 Marks]

[OR]

Q.4 (Alt) Explain finite word-length effects: coefficient quantization, roundoff noise, and limit cycles. [14 Marks]

--- UNIT V ---

Q.5 Describe multirate signal processing: Decimation and Interpolation by factor D and I. [14 Marks]

[OR]

Q.5 (Alt) Explain TMS320C67xx DSP processor architecture: Harvard bus, pipelining, and MAC unit. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****